

Mukesh Manoranjithan

Software Engineer | AI & Data Science Undergraduate | JLPT N3 Certified

+91-9952473896 manoranjithanmukesh@gmail.com

LinkedIn GitHub Portfolio

Professional Summary

Motivated Software Engineer and Artificial Intelligence Data Science undergraduate with expertise in Python, Java, SQL, and backend development. Experienced in developing full-stack applications, machine learning solutions, and AI-driven systems. JLPT N3 certified with strong problem-solving skills and a passion for building scalable software solutions. Recognized through AWS, IBM, NPTEL, and Forage certifications, along with two published patents.

Education

M. Kumarasamy College of Engineering, Karur

B.Tech in Artificial Intelligence and Data Science
2023 – 2027

Kongu Higher Secondary School

Higher Secondary (89%)

Technical Skills

Languages	Python, Java, SQL, C
Backend	PHP, REST APIs, MySQL
Frontend	HTML, CSS, JavaScript
Databases	MySQL, PostgreSQL
Machine Learning	Scikit-learn, Pandas, NumPy
Cloud	AWS Cloud, AWS Machine Learning Foundations
Developer Tools	Git, GitHub, Docker, VS Code, Jupyter Notebook
Concepts	Object-Oriented Programming, Data Structures, Algorithms, Database Management Systems, Operating Systems, Software Engineering

Projects

Personal Expense Tracker

GitHub

- Built a web-based expense management system using PHP, MySQL, HTML, CSS, and JavaScript.
- Implemented secure user authentication, expense categorization, budgeting, and transaction management.
- Designed an interactive dashboard for monthly expense summaries and financial insights.
- Optimized relational database queries for efficient data storage and reporting.

Technologies: PHP, MySQL, HTML, CSS, JavaScript, XAMPP

Insightful Indian Tourism Explorer

GitHub

- Developed a tourism analytics platform to analyze visitor trends and travel patterns across India.
- Performed data cleaning, transformation, and visualization using Python, SQL, and Power BI.
- Created interactive dashboards with KPIs, maps, and charts for data-driven decision making.
- Provided meaningful insights into tourism growth, seasonal demand, and regional performance.

Technologies: Python, SQL, Power BI, Pandas, Excel

Customer Churn Prediction and Business Insights

GitHub

- Developed a customer churn prediction model using Python, Pandas, NumPy, and Scikit-learn to identify customers at risk of leaving.
- Performed data preprocessing, feature engineering, exploratory data analysis (EDA), and model evaluation.
- Compared multiple machine learning algorithms using Accuracy, Precision, Recall, and F1-Score.
- Generated business insights through visualizations to support customer retention strategies.

Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib

Patents

BlurD – AI-Based Document Redaction System (Published Patent)

- Developed an intelligent document security system that automatically detects and redacts Personally Identifiable Information (PII) using Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), and Machine Learning techniques.
- Designed an automated redaction pipeline to identify sensitive entities such as names, addresses, identification numbers, and confidential organizational data while preserving document readability.
- Focused on improving document privacy, secure information sharing, and regulatory compliance for enterprise and government applications.
- Technologies: Python, RAG, NLP, Machine Learning, OCR, Document Processing.

Smart Helmet for Rider Safety (Published Patent)

- Designed and developed an IoT-enabled smart helmet capable of detecting road accidents and automatically transmitting emergency alerts with the rider's real-time GPS location.
- Integrated multiple sensors, GPS, GSM, and microcontroller-based communication modules to enable real-time monitoring and rapid emergency response.
- Improved rider safety by automating accident detection and minimizing emergency response time through intelligent notification mechanisms.
- Technologies: IoT, ESP32, GPS, GSM, Embedded Systems, Sensors.

Certifications

- AWS Educate Machine Learning Foundations – Amazon Web Services (2026)
- Japanese Language Proficiency Test (JLPT N3)
- Data Analytics and Visualization Job Simulation – Forage
- IBM Python 101 for Data Science
- IBM SQL and Relational Databases 101
- NPTEL Elite – Introduction to Cognitive Psychology (79%)
- NPTEL – The Joy of Computing using Python
- NPTEL – Introduction to Industry 4.0 and Industrial Internet of Things

Languages

English	Professional Working Proficiency
Tamil	Native
Japanese	JLPT N3 Certified